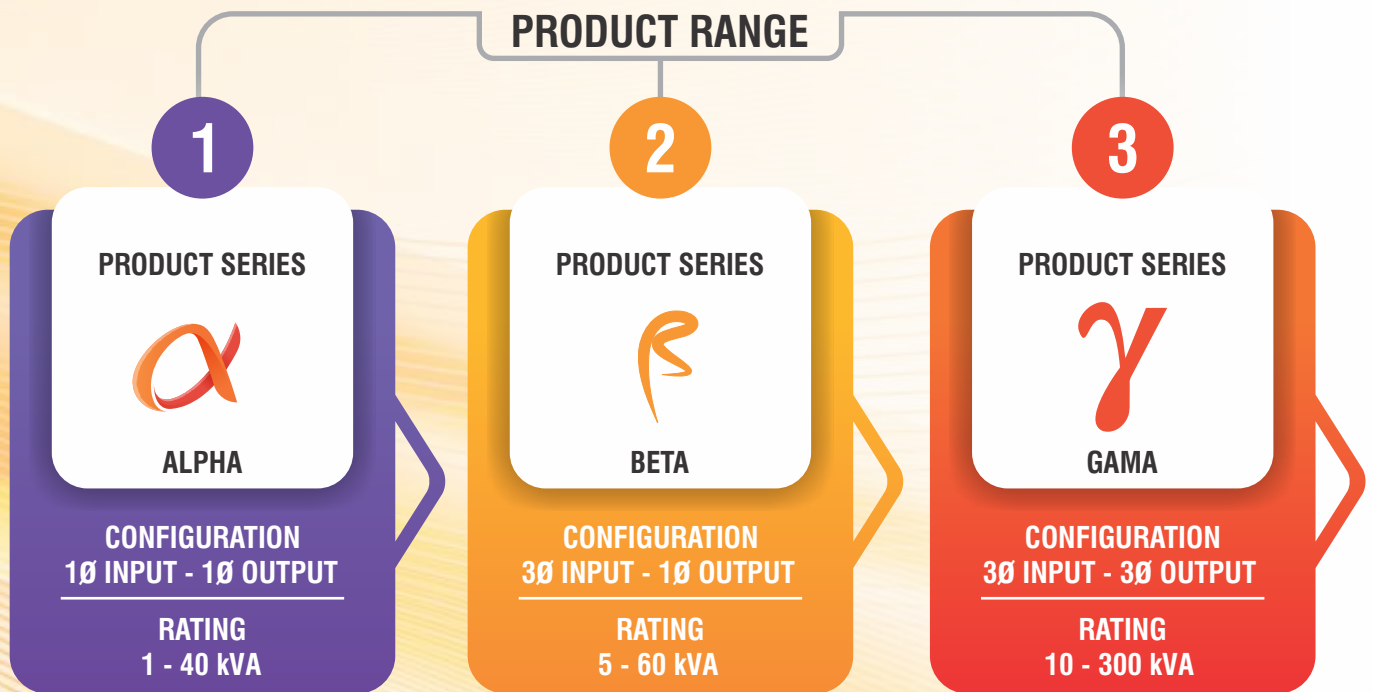


SHIVI ELECTRONICS SYSTEM



ONLINE UPS 1kVA – 300kVA



FEATURES

DSP BASED DOUBLE CONVERSION TOPOLOGY

At input stage AC supply is converted to controlled DC and in second stage DC is converted to controlled AC which is used to support the critical loads. Due to double conversion the load is completely protected against spikes, surges and brownouts. Use of DSP ensures low component count and reliability of the circuit.

HIGH FREQUENCY PWM BASED INVERTER USING IGBTs

Conversion from DC to controlled and pollution free AC is achieved through high frequency PWM based Inverter using IGBTs with instantaneous sinewave control. The technology offers following benefits to the users.

- High efficiency
- Low noise
- Compact size
- Faster transient response

INDEPENDENT LCD

In RM series each Power Module has independent LCD which displays Status and Alarms in real time.

PROTECTION AGAINST DC SHOOT THROUGH FAULT

Possibilities of DC supply getting short circuited due to "both top & bottom IGBTs getting switched on simultaneously". In such situation the devices may get damaged due to flow of very high current through IGBTs caused due to sudden discharge of DC link capacitors. To prevent IGBTs from getting damaged in this process UPS controller has foolproof DC shoot through fault protection. As a result in case of DC shoot through fault occurrence, the protection will instantly switch off the IGBT & protect it.



ALPHA SERIES

1Ø INPUT - 1Ø OUTPUT
1 - 40 kVA



Standard Features:

- Wide Input Voltage Range
- High Input Power Factor
- Protection against Over Voltage, Short Circuit, and Over Temperature
- Auto Fan Speed Adjustment
- LCD Display Monitoring-all the operation status

Optional Features:

- Dry Contacts
- Battery Cold Start
- Static Bypass

INPUT

Nominal Voltage	230VAC
Voltage Range	230 $\pm$ 20%
Frequency	50Hz

OUTPUT

Voltage	220/230/240 (Selectable)
Voltage Regulation	1%
Total Harmonic Distortion	2%
Power Factor	0.9
Crest Factor	3:1
Overload (Inverter Mode)	110% for 2 minute

DC LINK

Battery Voltage	120, 144, 180, 192VDC
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SYSTEM

Efficiency (Inverter)	upto 92% - 95%
Display	LED + LCD
Degree of Prediction	Ip20
Audible Noise	50db

ENVIRONMENTAL CONDITIONS

Operating Temperature	0-40°C
Storage Temperature	-20°C to 70°C
Relative Humidity	0-95% (Non-Condensing)
Maximum Altitude above Sea Level	1000 Meters without derating



BETA SERIES

3Ø INPUT - 1Ø OUTPUT
5 - 60 kVA



Standard Features:

- High Frequency PWM Using IGBTs
- In-built Isolation Transformer
- Auto Re-transfer Static Switch
- Manual Bypass
- Extended Battery Back-up

INPUT

Nominal Voltage	3Ø, 380 / 400 / 415V
Voltage Range	350 to 460
Frequency	50Hz
Power Factor	0.99 (Optional)

OUTPUT

Nominal Voltage	220/230/240 (Selectable)
Frequency	50Hz
Power Factor	0.9
Voltage Regulation	$\pm$ 1%
Voltage Distortion	$\leq$ 3%
Overload	110% for 2 minute
Crest Factor	3:1
Efficiency (Inverter)	upto 92% - 95%

DC LINK

Battery Voltage	240V, 360VDC
Voltage Tolerance	$\pm$ 1%
Voltage Ripple	$\pm$ 1%

STATIC BYPASS (OPTIONAL)

Switching Time	Inverter Failure: 2ms, Other modes 0
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ENVIRONMENTAL CONDITIONS

Operating Temperature	0-40°C
Storage Temperature	-20°C to 70°C
Relative Humidity	0-95% (Non-Condensing)
Maximum Altitude above Sea Level	1000 Meters without derating

GENERAL

Ingress protection	Ip20
Audible Noise at 1 Mtr	< 60dBA
Cooling	Forced Air

GAMA SERIES **3Ø INPUT - 3Ø OUTPUT** **10 - 300 kVA**



Standard Features:

- High Frequency PWM Using IGBTs
- In-built Isolation Transformer
- Auto Re-transfer Static Switch
- Energy Saving Mode
- Parallel Redundancy
- Protection against DC Shoot through Fault

Optional Features:

- Remote Monitoring for Alarms/Indications
- Remote Emergency Power Off
- Potential Free Contacts for Remote Indications
- Voltage Stabiliser in Bypass Line

INPUT	
Nominal Voltage	3Ø, 380 / 400 / 415V
Voltage Range	350 to 460
Frequency	50Hz or 60Hz (Selectable)
Frequency Variation	40Hz or 60Hz
OUTPUT	
Nominal Voltage	220/230/240 (Selectable)
Frequency	50Hz or 60Hz
Frequency Synchronization Range	±1%
Power Factor	0.9
Voltage Regulation	Balanced Load $\pm 1\%$
Phase Displacement	Balanced Load $120^\circ \pm 1\%$
Voltage Distortion	$\leq 3\%$
Overload	125% for 10 minute
Crest Factor	3:1
Efficiency (Inverter)	upto 93%
Power Factor	0.9
Voltage Regulation	$\pm 1\%$
DC LINK	
Battery Voltage	360V, 384 Volts
Voltage Regulation	$\pm 1\%$
Voltage Ripple	$\pm 1\%$
ENVIRONMENTAL CONDITIONS	
Operating Temperature	0-40°C
Storage Temperature	-20°C to 70°C
Relative Humidity	95% (Non-Condensing)
Maximum Altitude above Sea Level	1000 Meters without derating
GENERAL	
Ingress protection	IP20
Cooling	Forced Air



SHIVI ELECTRONICS SYSTEM

Corp. Office: 734, Tulsi Vihar, Dadri 203207

Branch Office: Khas 104M, Khera Chauganpur,

Greater Noida, G B Nagar - 201306

Email: shivielelectronicsyst@gmail.com | Website: shivielelectronicsyst.com